

ONLINE QUIZ APPLICATION

Java Swing Desktop Application – Project Report

PROJECT REPORT

Abhishek Vashishtha
B.Tech CSE (AI & ML)
VIT Bhopal University

1. Introduction

The Online Quiz Application is a Java-based desktop application developed using Core Java and Java Swing. It provides an interactive platform where a user can enter a name, attempt multiple-choice questions, receive immediate answer validation, and view the final score, percentage, and grade. The application is designed as a simple, executable academic project that demonstrates Java programming and Object-Oriented Programming concepts.

2. Objectives

- Develop an interactive GUI-based quiz application.
- Provide multiple-choice Java programming questions.
- Implement a countdown timer for each question.
- Automatically evaluate submitted answers.
- Calculate score, percentage, and grade.
- Demonstrate practical use of Java OOP and Swing.

3. Technologies Used

Programming Language: Java

GUI Framework: Java Swing

Concepts: OOP, Classes and Objects, ArrayList, Event Handling, Exception-safe input flow

Development Environment: VS Code / any Java-compatible IDE

Database: Not required

External Libraries: Not required

4. Main Features

- Student/User name entry
- 20 Java MCQ question bank
- Randomized question order
- 15 questions selected for each quiz
- Four options per question
- 30-second timer for each question
- Next and Submit controls
- Automatic score calculation
- Percentage calculation
- Grade generation
- Restart Quiz and Exit options

5. Project Structure

Main.java - Starts the application.

Question.java - Stores question text, options, and correct answer.

QuizManager.java - Maintains the question bank and randomizes questions.

LoginFrame.java - Provides the user-name screen.

QuizFrame.java - Displays questions, options, timer, and quiz controls.

ResultFrame.java - Displays score, percentage, and grade.

README.md - Contains project documentation and execution instructions.

6. Working of the System

1. The application starts from Main.java.
2. The Login Frame asks the user to enter a name.
3. After clicking Start Quiz, the Quiz Frame opens.
4. A set of 15 questions is selected from the 20-question Java question bank.
5. Each question provides four options and a 30-second countdown.
6. The selected answer is checked when the user moves to the next question or submits the quiz.
7. At the end, the Result Frame calculates and displays the score, percentage, and grade.
8. The user can restart the quiz or exit the application.

7. OOP Concepts Used

Encapsulation: Question data and related behavior are organized inside classes.

Abstraction: Quiz management and GUI operations are separated into dedicated classes.

Objects and Classes: Question, QuizManager, LoginFrame, QuizFrame, and ResultFrame are implemented as classes/objects.

Modularity: Different responsibilities are divided into separate Java files, making the project easier to maintain.

8. Grading System

A+ : 90% and above

A : 80%-89%

B : 70%-79%

C : 60%-69%

D : 50%-59%

F : Below 50%

9. Project Executability

The project is a fully executable Java desktop application. It requires JDK 8 or later and can be executed in VS Code or another Java IDE. No database or external dependency is required.

Terminal execution:

1. Open the project folder.
2. Compile using: `javac *.java`
3. Run using: `java Main`

The graphical user interface opens after the application starts.

10. Testing

The application can be tested by entering a valid user name, starting the quiz, selecting answers, observing the countdown timer, moving through questions, submitting the quiz, and checking the result screen. Restart and exit functions can also be verified.

11. Advantages

- Simple and user-friendly interface
- Automatic evaluation
- Time-bound quiz experience
- Easy to execute and demonstrate

- Uses standard Java libraries
- Clear separation of project components

12. Future Enhancements

- Add multiple categories such as C++, Python, and General Knowledge
- Add a larger question database
- Add difficulty levels
- Store scores in a database
- Add a leaderboard
- Add user authentication
- Add sound effects and improved visual themes
- Generate detailed performance reports

13. Conclusion

The Online Quiz Application successfully demonstrates how Java can be used to build an interactive desktop application. It combines Swing-based GUI development, Object-Oriented Programming, collections, event handling, timers, and result processing in one practical project. The project is suitable for academic demonstration and can be extended with additional features in the future.

14. Author

Abhishek Vashishtha
B.Tech CSE (AI & ML)
VIT Bhopal University